

3(a)(i)	D (nitrogenous organic / nucleotide) base ; A pyrimidine / uracil / cytosine R purine E ribose ; I pentose F phosphate (group) ;	3
3(a)(ii)	<i>any one from</i> two, strands / polynucleotides (not one) ; A two chains R polypeptides deoxyribose (not ribose) ; A correct ref. to –OH (on C2) instead of –H group on the pentose / sugar <u>thymine</u> (not uracil) ; R thiamine have base pairs ; (double) helix / helical (not straight chain) ; longer ;	1
3(b)	<i>any four from</i> 1 template RNA binds to (part of) region X ; 2 in <u>active site</u> of, telomerase / enzyme ; 3 DNA / free, nucleotides, pair with / bind to / align with, RNA ; 4 ref. to complementary, bases / base pairs ; 5 A–T and C–G and U–A ; 6 phosphodiester bonds form, between (DNA) nucleotides / AW ; treat as neutral any <i>refs.</i> to likely enzymes, e.g. ligase, etc. 7 telomerase moves, in the direction of the arrow / to the right ; 8 AVP ; e.g. hydrogen bonds between bases telomerase acts as a reverse transcriptase	4

Question	Answer	Marks
3(c)	<i>prokaryotes</i> have, circular DNA ; A 'loop of DNA' I plasmid(s) do not have telomeres ; do not have chromosomes ; AVP ; e.g. 'DNA has no ends'	1
3(d)	<i>any two from</i> cancer cells divide, uncontrollable / continuously ; ORA A cancer cells divide but do not undergo cell death telomerase is required to ensure, <u>chromosomes</u>, do not shorten ; telomerase is, synthesised / produced / found, in cancer cells ; ORA high levels of telomerase indicate cells that are cancerous / AW ; AVP ; e.g. <i>idea that</i> gene for telomerase switched on	2